

Tom Zhang, Cert CII (London Market)

pnlync@gmail.com · linkedin.com/in/zhang42 · Dublin, Ireland

Actuarial Projects: pnlync.github.io/actuarial

Profile

Mathematics graduate with an MSc in Cloud Computing and hands-on actuarial project work spanning pricing, valuation, reserving and capital modelling. Eligible to work without employer sponsorship.

Education

National College of Ireland

Master of Science in Cloud Computing

Dublin, Ireland

Sep. 2025 – Sep. 2026

Sun Yat-sen University

Bachelor of Science in Mathematics (Information and Computing Science)

Guangdong, China

Sep. 2020 – Jun. 2024

Professional Qualifications

Institute and Faculty of Actuaries (IFoA)

- CS1 Actuarial Statistics
- CB2 Business Economics
- CB3 Business Management
- Stage 1 and Stage 2 Professionalism Courses
- CM1 Actuarial Mathematics for Modelling — Results pending
- CS2 Risk Modelling and Survival Analysis — Results pending

Chartered Insurance Institute (CII)

Certificate in London Market Insurance

- LM1 London Market Insurance Essentials
- LM2 London Market Insurance Principles and Practices
- LM3 London Market Underwriting Principles

Projects

Term Life Pricing, Valuation & Capital Model

pnlync.github.io/assurance

Life Insurance | *IFRS 17, Solvency II, Profit Testing, Experience Studies, Analysis of Change*

- Built one seriatim cash-flow engine for a synthetic 50K-policy Irish term life and mortgage protection book, driving pricing, IFRS 17 and Solvency II, and reconciled it to the cent against an independent Excel model
- Priced unisex rates to a 10% margin by profit testing, showing the €60 policy fee left small covers loss-making (break-even €177K at age 35), the same cells IFRS 17 grouped as onerous
- Ran A/E experience analysis with credibility on 3 simulated years, adopting the lapse signal in full but only a quarter of the mortality signal; the new basis cut CSM by 16% and required 2–8% repricing
- Valued the book under IFRS 17 (Monte Carlo RA, CSM roll-forward) and Solvency II (BEL, SCR), explained each year's movements in an analysis of change, and quantified the EU 2027 risk-margin reform at –36%

Defined Benefit Pension Valuation, Funding & De-risking Model

pnlync.github.io/db-pension

Retirement | *IAS 19, Irish Funding Standard & FSR, AoC, LDI, Buy-in Pricing*

- Valued a synthetic 1,000-member Irish final-salary scheme under IAS 19 (€206.8M) and the Funding Standard (€203.0M), bridging the €3.8M gap: –€18.8M on non-pensioners, +€15.1M for annuities and wind-up
- Rolled the scheme through 2025 on actual yields, MVAs and asset returns with no unexplained residual: rising rates cut the IAS 19 deficit from €33.2M to €2.1M, yet it stayed €11.5M short of FS + FSR
- Sized a 3-year funding plan at €4.5M a year; a 20% equity-to-sovereign switch cut the FS + FSR shortfall by 64% and lifted the IAS 19 hedge ratio from 32% to 52%, at €1.3M a year of expected return
- Priced a pensioner buy-in from an insurer's Solvency II view at 117% of its IAS 19 value (€19.9M OCI loss under exact-match accounting), and found the quote 7% above the statutory annuity basis

Return-Shipping Insurance Pricing Engine for E-commerce pnlync.github.io/shipping-insurance

General Insurance | *GLM, Credibility, XGBoost, Loss Ratio Monitoring, Stress Testing*

- Built a frequency-severity GLM pricing cover to a 60% target loss ratio on 110K real order items with simulated claims, using only quote-time features and matching held-out losses within 0.3%
- Credibility-weighted the sparse experience of 2,970 sellers against the GLM rate at unchanged total premium, cutting the riskiest tier's A/E from 1.56 to 1.36 and flagging it for watch at an 82% loss ratio
- Kept the GLM as champion after an XGBoost challenger ranked risk barely better but missed held-out losses by 1.5% (GLM 0.3%); a +20% frequency and severity stress lifts the loss ratio to 85.5%

Commercial Auto IBNR Estimation & Blind Back-Test pnlync.github.io/reserving

Reserving | *Chain Ladder, Bornhuetter-Ferguson, Cape Cod, Benktander, Mack, Bootstrap*

- Estimated year-end 2007 reserves for a real US commercial auto insurer from NAIC Schedule P claims triangles using chain ladder, BF and Cape Cod: \$365.4M, 44% above what the insurer had booked
- Tested the methods on earlier year-ends first, switched mature years to incurred chain ladder after paid chain ladder under-projected, and locked the selection in git before looking at any later data
- Scored it against nine years of actual run-off: the projection fell 10.3% short of actual payments (pure chain ladder 15.6%); on 120-month incurred it was 11.2% low vs 38.6% for the insurer's booked reserve
- Measured reserve uncertainty with Mack and a bootstrap; rerun on 45 insurers, the model's 75th percentile covered the actual outcome for only two-thirds of them, so its ranges were too narrow

Flood Catastrophe Reinsurance & Capital Optimisation pnlync.github.io/reinsurance

Reinsurance | *Cat XoL, Reinstatements, OEP/AEP, Economic Capital, Stress Testing*

- Built an event-level flood catastrophe model from public NFIP claims and policy data, restating 20 Gulf Coast events from 2009-23 onto \$805B of 2025 insured value and simulating 100,000 years of losses
- Structured quota share and Cat XoL towers with coinsurance and reinstatements, and priced each layer on modelled loss plus a diversified capital load, from $2.2\times$ expected loss at 1-in-5 to $8.1\times$ at 1-in-100
- Compared 255 programmes on a cost-capital frontier: the recommended one cut 1-in-200 economic capital by 81.7% for \$1.2B a year in net cost, an implied 5.9% cost of capital against a 10% hurdle
- Stress-tested the choice: its layer structure held under all 9 shocks, but removing Hurricane Harvey cut gross capital by 57%, and without reinsurer diversification the same rule bought no reinsurance

Micro-Certificates and Courses

OpenAI Academy

- Applied AI Foundations
- Agents and Workflows

Claude Academy

- AI Fluency: Framework & Foundations
- AI Capabilities and Limitations

Casualty Actuarial Society

- CAS P&C Actuarial Essentials Program

Open Actuarial Textbooks Project

- Loss Data Analytics
- Life Contingencies

Technical Skills

Pricing: Profit Testing, Frequency-Severity GLMs, Credibility, XGBoost, Loss Ratio Monitoring

Valuation & Capital: IFRS 17 / HKFRS 17 (GMM, CSM, RA), Solvency II (BEL, SCR, Risk Margin), Analysis of Change

Pensions: IAS 19 (PUC, DBO), Irish Funding Standard & FSR, Transfer Values, PV01 & LDI, Buy-in Pricing

Experience & Risk: A/E Experience Studies, Assumption Setting, Stress Testing, Monte Carlo, Model Validation

Reserving: Chain Ladder, Bornhuetter-Ferguson, Cape Cod, IBNR, Mack, Bootstrap, Back-testing

Reinsurance: Quota Share, Cat XoL, Reinstatements, OEP/AEP, Burning Cost, Economic Capital

Programming: Python (pandas, NumPy, SciPy), SQL, R, Excel, Git, pytest, AWS, AI Agents (Claude Code, Codex)